

THERMAL IMAGING INSPECTION REPORT

Site: Greenfield Residential Complex, Block C — Pune, Maharashtra

Report No: TSI-THR-2024-0312 | Date: 11 April 2025

1. Survey Details

Client	BuildSafe Engineering Pvt. Ltd. (on behalf of Greenfield Developers)
Site	Block C, Greenfield Residential Complex, Baner Road, Pune – 411045
Survey Date	10 April 2025, 06:00–09:30 hrs (pre-dawn survey)
Ambient Temperature	24°C (survey start) — 29°C (survey end)
Relative Humidity	68% RH
Wind Speed	< 2 m/s (calm conditions — suitable for survey)
Camera Model	FLIR E96 — Resolution: 640 x 480 IR, FOV 24 degrees
Emissivity Setting	0.95 (concrete/masonry surfaces)
Thermographer	Mr. Vivek Kulkarni, Certified Level-II Thermographer (ISO 18436-7)
Report Reference	TSI-THR-2024-0312

2. Methodology

Thermal imaging was conducted during the pre-dawn window (06:00–09:30 hrs IST) to exploit maximum delta-T between the building envelope and ambient air. Passive thermography technique was used — no active heating was applied. All images were captured in radiometric JPEG format and processed using FLIR Tools+ software. Areas of interest were cross-referenced with the concurrent civil visual inspection (Report BSE-INS-2024-0471).

Temperature thresholds for anomaly classification: Delta-T < 2°C — No anomaly; Delta-T 2–5°C — Possible anomaly (monitor); Delta-T 5–10°C — Probable moisture or void; Delta-T > 10°C — Active moisture / significant thermal bridge.

3. Thermal Findings by Zone

Zone T-01: Terrace Slab — Central Bay

Image Ref	IMG_THR_001 to IMG_THR_007
Surface Temp (avg)	26.4°C
Anomaly Temp (max)	34.8°C (delta-T: +8.4°C above background)

Pattern	Large irregular warm patch 3.2 m x 2.1 m at centre of slab indicating trapped moisture beneath screed layer.
Classification	Probable moisture entrapment — active seepage zone.
Confidence	High (corroborated by visual seepage at flat 401 ceiling)

Zone T-02: Terrace Slab — North Parapet Junction

Image Ref	IMG_THR_008 to IMG_THR_011
Surface Temp (avg)	25.9°C
Anomaly Temp (max)	31.2°C (delta-T: +5.3°C)
Pattern	Linear warm band along parapet-slab junction approx. 4.5 m length; consistent with water infiltration at parapet base flashing failure.
Classification	Probable moisture — linear seepage path.
Confidence	Medium-High

Zone T-03: East Facade — 2nd Floor Window Band

Image Ref	IMG_THR_012 to IMG_THR_016
Surface Temp (avg)	25.2°C
Anomaly Temp (max)	28.7°C (delta-T: +3.5°C)
Pattern	Diffuse warm areas around 6 window reveals; moisture tracking pattern from sill to soffit. Pattern consistent with failed sill waterproofing.
Classification	Possible moisture — window reveal infiltration.
Confidence	Medium

Zone T-04: External Wall — Expansion Joint at 2nd Floor

Image Ref	IMG_THR_017 to IMG_THR_019
Surface Temp (avg)	25.5°C
Anomaly Temp (max)	32.4°C (delta-T: +6.9°C)
Pattern	Continuous warm stripe at expansion joint level, 11 m length across full width of east facade. Matches failed sealant observed in visual inspection.
Classification	Probable moisture — open expansion joint.

Confidence	High
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Zone T-05: Parking Podium Soffit — Beam Grid

Image Ref	IMG_THR_020 to IMG_THR_026
Surface Temp (avg)	24.8°C
Anomaly Temp (max)	27.6°C (delta-T: +2.8°C)
Pattern	Mild warm patches at 4 beam soffit locations; pattern suggestive of moisture-induced concrete delamination rather than active seepage. No high delta-T indicating large water pockets.
Classification	Possible thermal anomaly — delamination or residual moisture.
Confidence	Medium

Zone T-06: Staircase Soffit G to 1st Floor

Image Ref	IMG_THR_027 to IMG_THR_028
Surface Temp (avg)	25.1°C
Anomaly Temp (max)	25.9°C (delta-T: +0.8°C)
Pattern	No significant thermal anomaly detected. Temperature variation within acceptable limits.
Classification	No anomaly.
Confidence	High

4. Summary of Thermal Anomalies

Zone	Location	Delta-T (°C)	Classification	Priority
T-01	Terrace — Central Bay	+8.4	Active Moisture	HIGH
T-02	Terrace — N Parapet Jn.	+5.3	Moisture Seepage	HIGH
T-03	East Facade — Windows	+3.5	Moisture Possible	MEDIUM
T-04	Expansion Joint E-Facade	+6.9	Active Moisture	HIGH
T-05	Parking Beam Soffit	+2.8	Possible Delamination	MEDIUM
T-06	Staircase Soffit	+0.8	No Anomaly	LOW

5. Thermal Recommendations

- Zones T-01 and T-02 (Terrace) require immediate moisture remediation. Thermal imaging confirms extent of moisture entrapment — full waterproofing overhaul is warranted as recommended in the accompanying civil report.
- Zone T-04 (Expansion Joint) — sealant replacement is urgent; thermal evidence shows active moisture penetration over 11 m stretch.
- Zone T-03 (Window Reveals) — apply silicone sealant bead around all window perimeters and replace damaged sill drip profiles.
- Zone T-05 (Parking Soffit) — recommend follow-up core-cut or impact-echo testing to differentiate delamination from residual moisture. Low priority if no structural concern confirmed by structural engineer.
- Repeat thermal survey 3 months after remediation works to verify effectiveness of repairs. Survey to be conducted under similar pre-dawn conditions.

6. Limitations

Thermal imaging is a non-destructive survey technique. Results are indicative and must be corroborated with invasive investigation where required. Results may vary with changing ambient conditions. Internal areas not accessible at the time of survey are not covered by this report. This report is valid only for the conditions prevailing on the survey date.

Thermographer: Mr. Vivek Kulkarni, Level-II (ISO 18436-7)

Reviewed by: Ms. Priya Nair, Technical Director — ThermoScan India Pvt. Ltd.

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